

General Knowledge

Indian Geography

STUDY NOTES

Dams in India



Dams in India

A dam is a type of barrier designed to store water; the resultant reservoir is used for irrigation, generating electricity, etc.

Dams are categorized according to structure, intended objective, etc.

- **By Structure-** Embankment dams, Arch dams, Gravity dams, etc.
- **By Purpose-** Power generation, Irrigation purpose, Multipurpose, etc.

- There is a lot of speculation about the origin of dams, some strong pieces of evidence in history point out to the Jawa Dam which was constructed in 3000 BCE
- After the success of the Jawa Dam, Egyptians built the Sadd el-Kafara, or Dam of the Pagans, which was mainly for irrigation purposes.
- In the current era, the Quatinah Barrage or Lake Homs Dam, located in Syria, is the oldest operational dam in the world.
- In India, the oldest dam is the Grand Anicut Dam or Kallanai Dam which was built on the Cauvery River by King Karikalan of the Chola dynasty in the first century.
- This dam was built with uneven stones and stands at a length of 329 meters and a width of 20 meters.
- The main intention of this dam was to divert the water across the delta for irrigation purposes.



List of Dams in India with River and State

State	Dam	River
Andhra Pradesh	Somasila Dam	Penna River

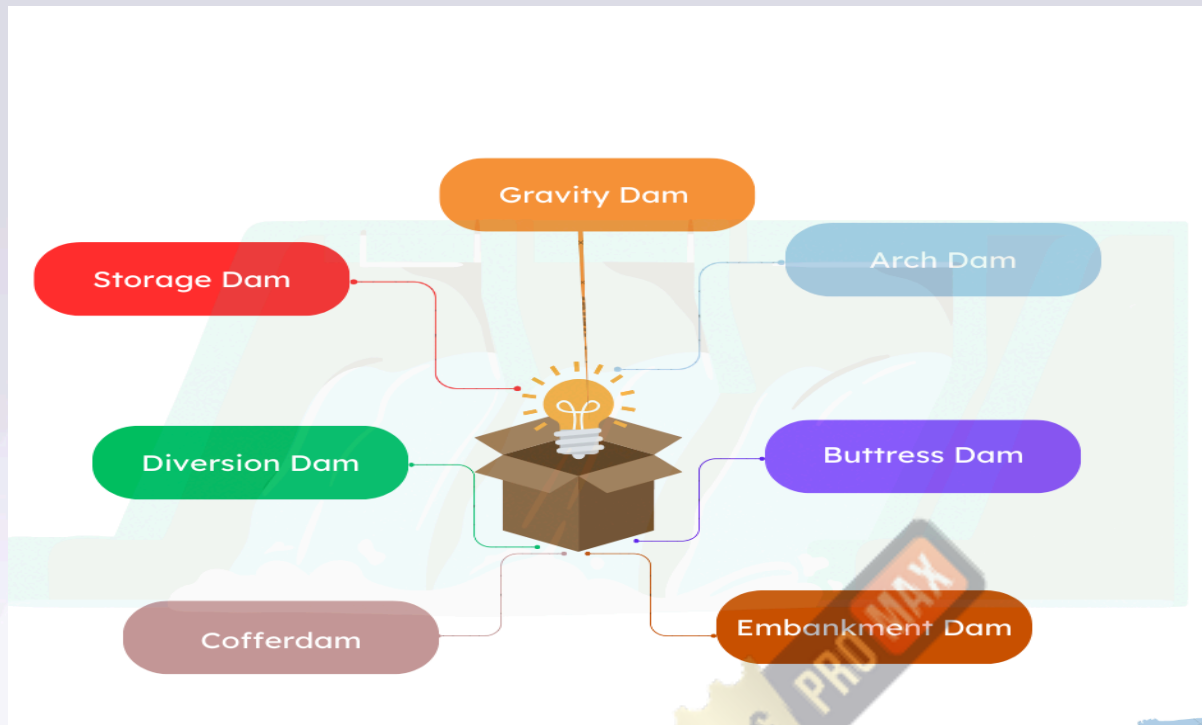
	Polavaram Project	Godavari River
	Nagarjuna Sagar Dam	Krishna River
	Srisaillam Dam	Krishna River
Arunachal Pradesh	Kohira Dam	Kohira River
Bihar	Nagi Dam	Nagi River
	Ranganadi Dam	Ranganadi River
Chhattisgarh	Dudhawa Dam	Mahanadi River
	Minimata (Hasdeo) Bango Dam	Hasdeo River
Gujarat	Aji Dam	Aji River
	Sipu Dam	Sipu River
	Mitti Dam	Mitti River
	Karjan Dam	Karjan River
	Kadana Dam	Mahi River
	Sardar Sarovar Dam	Narmada River
	Ukai Dam	Tapti River
Himachal Pradesh	Chamera Dam	Ravi River
	Nathpa Jhakri (Sjvnl) Dam	Satluj River
	Pong Dam	Beas River
	Bhakra Dam	Satluj River
	Kol Dam	Satluj River
Jammu and Kashmir	Pakal Dul Dam	Marusudar River
	Kishenganga Dam	Kishenganga River
	Nimoo Bazgo Dam	Indus River
	Uri Dam	Jhelum River
	Salal Dam	Chenab River
	Baglihar Dam	Chenab River

Jharkhand	Konar Dam	Konar River
	North Koel Dam	North Koel River
	Tenughat Dam	Damodar River
	Maithon Dam	Barakar River
	Panchet Dam	Damodar River
Karnataka	Lakhya Dam	Lakhya Hole
	Tungabhadra Dam	Tungabhadra River
	Almatti Dam	Krishna River
	Kabini Dam	Kabini River
	Supa Dam	Kalinadi River
	Hidkal Dam	Ghataprabha River
	Hemavathy Dam	Hemavathy River
	Bhadra Dam	Bhadra River
	Basava Sagar Dam (Narayanpur Dam)	Krishna River
	Krishnarajasagar Dam	Cauvery River
Kerala	Idukki Dam	Periyar River
	Cheruthoni Dam	Cheruthoni River
	Kakki Dam	Kakki River
	Kulamavu Dam	Kilivillithode
	Mullaperiyar Dam	Periyar River
Madhya Pradesh	Ban Sagar Dam	Son River
	Gandhi Sagar Dam	Chambal River
	Indira Sagar Dam	Narmada River
	Omkareshwar Dam	Narmada River
	Tawa Dam	Tawa River
Maharashtra	Bhatsa Dam	Bhatsa and Chorna Rivers

	Koyna Dam	Koyna River
	Warna Dam	Warna River
	Ujjani Dam	Bhima River
	Aruna Dam	Aruna River
	Upper Wardha Dam	Wardha River
Odisha	Hirakud Dam	Mahanadi
	Indravati Dam	Indravati
	Kapur Dam	Kapur
	Podagada Dam	Podagada
	Rengali Dam	Brahmani
	Upper Kolab Dam	Kapur
	Haladia Dam	Haladia
	Lower Indra Dam	Indra
Punjab	Ranjit Sagar Dam	Ravi
Rajasthan	Bisalpur Dam	Banas
	Jawahar Sagar Dam	Chambal
	Mahi Bajaj Sagar Dam	Mahi
	Rana Pratap Sagar Dam	Chambal
	Jaswant Sagar Dam	Luni
	Jakham Main Dam	Jakham (Mahi)
Sikkim	Rangit III Dam	Ranjit
Tamil Nadu	Bhavani Dam	Bhavani
	Mettur Dam	Kaveri
	Sholaiyar Dam	Sholaiyar
	Pillur Dam	Bhavani
Telangana	Nagarjuna Sagar Dam	Krishna

	Srisaillam Dam	Krishna
	Nizam Sagar Dam	Manjira
	Musi Dam	Musi
	Singur Dam	Manjira
	Sri Rama Sagar Dam (Pochampadu Project)	Godavari
Uttarakhand	Jamrani Dam	Gola
	Lakhwar Dam	Yamuna
	Koteswar Dam	Bhagirathi
	Ramganga Dam	Ramganga
	Tehri Dam	Bhagirathi
Uttar Pradesh	Rihand Dam	Rihand
West Bengal	Kangsabati Kumari Dam	Kasai

Types of Dams



➤ **Gravity Dam:**

- Description: These dams are heavy and massive, using their own weight to resist the force of water.
- Example: Bhakra Dam in India.

➤ **Arch Dam:**

- Description: Curved in shape, these dams transfer water pressure to the valley walls.
- Example: Idukki Dam in India.

➤ **Buttress Dam:**

- Description: Supported by multiple triangular-shaped walls (buttresses) on the downstream side.
- Example: Tehri Dam in India.

➤ **Embankment Dam:**

- Description: Made from compacted earth (earthfill) or rock (rockfill). These are the most common types of dams.
- Example: Hirakud Dam in India (earth fill).

➤ **Cofferdam:**

- Description: Temporary structures designed to keep the water out of the construction area.
- Example: Used during the construction of permanent dams.

➤ **Diversion Dam:**

- Description: Redirects the flow of a river or stream into a different channel.
- Example: Sardar Sarovar Dam in India.

➤ **Storage Dam:**

- Description: Created primarily to store water for irrigation, water supply, and hydroelectric power.
- Example: Nagarjuna Sagar Dam in India.

Interesting Facts about Dams in India

Fact	Dam	Description	Image
First dam in India	Kallanai Dam (Grand Anicut)	Located on the river Kaveri in Tiruchirapalli, Tamil Nadu. Built around the 2nd century AD.	
Longest dam in the world	Hirakud Dam	Located in Odisha, it stretches across 25.8 km.	
Longest dam in India	Hirakud Dam	The same as above; it holds the record in both categories.	
Highest straight gravity dam in India	Bhakra Dam	Located on the Sutlej River in Himachal Pradesh, standing 225 meters high.	
Tallest dam in the world	Nurek Dam	Situated in Tajikistan, with a height of 300 meters.	

Highest dam in India	Tehri Dam	Located in Uttarakhand, it stands at 260.5 meters high.	
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Tehri Dam

The Tehri Dam is located in the state of Uttarakhand. It is the **highest dam** in India with a height of 260.5 metres. It is situated on the river Bhagirathi.

- **Height of the Dam-** 260.5 m
- **Length of the Dam-** 575 m
- **Type of Dam-** Rock fill
- **Reservoir capacity-** 21,00,000 acre feet
- **Installed capacity -** 1000 MW

Bhakra Nangal Dam

The Bhakra Nangal Dam is located in the states of Himachal Pradesh and Punjab. It is the **largest dam** in India having a height of 225 metres. It is situated on the river Sutlej.

- **Height of the Dam-** 226 m
- **Length of the Dam-** 520m
- **Type of Dam-** Concrete Gravity
- **Reservoir capacity-** 75,01,775 acre feet
- **Installed capacity-** 1325 MW

Hirakud Dam

The Hirakud Dam is located in the state of Orissa. It is the **longest dam** in India with a total length of 25.79 km. It is situated on the river Mahanadi.

- **Height of the Dam-** 61 m
- **Length of the Dam-** 4.8 km (Main Dam)
- **Type of Dam-** Composite Dam
- **Reservoir capacity-** 47,79,965 acre feet
- **Installed capacity-** 347.5 MW

Nagarjuna Sagar Dam

The Nagarjuna Sagar Dam is located in the state of Telangana. It is **India's largest masonry dam** built till date. It is the **largest manmade lake** in the world. It has 26 gates and is 1.55 km in length. It is situated on the river Krishna.

- **Height of the Dam-** 124m
- **Length of the Dam-** 4.863 km (Total Length)
- **Type of Dam-** Masonry Dam
- **Reservoir capacity-** 93,71,845 acre feet
- **Installed capacity-** 816 MW

Sardar Sarovar Dam

The Sardar Sarovar Dam is located in the state of Gujarat. It is the **largest dam in the Narmada Valley Project**. This dam was built to benefit the neighbouring states of Madhya Pradesh, Rajasthan and Maharashtra. It is situated on the Narmada river.

- **Height of the Dam-** 163m
- **Length of the Dam-** 1210m
- **Type of Dam-** Gravity Dam
- **Reservoir capacity-** 77,00,000 acre-feet
- **Installed capacity-** 1450 MW

Indiranagar Dam

The Indira Sagar Dam is a multipurpose concrete gravity dam in Madhya Pradesh over the Narmada River in the town of Narmada Nagar. The dam contains the **largest reservoir** in India in terms of water storage capacity.

- **Height of the dam-** 92m
- **Length of the dam-** 653m
- **Type of dam-** Gravity Dam
- **Reservoir capacity-** 12.2 billion cu.m
- **Installed capacity:** 1000 MW